

OPERATING SYSTEMS: SYNCRORIZATION PRIMITIVES CONCEPTS & BASIC MECHANISMS

ADVANCED TOOLS

⌚ CORE CONCEPTS

- 1. Race Condition:** When multiple threads access resources and depends on the order of execution.
- 2. Critical Section**
Sections where shared resources are accessed.
- 3. Critical Section**
Guaranteeing that only one thread inside the execution of a line.
- 3. Mutual Exclusion**
- 2. Condition Signal**
Wakes up only ONE waiting thread. Operates with one-to-one broadcast.
- 4. Producer-Consumer Problem**
Classic example solved with a bounded buffer.

- 1. Locks (Mutexes)**
Simple binary (0/1) mechanism. acquire() to enter CS, release() to leave CS. Ensure mutual exclusion.
- 2. Semaphores**
Integer variable with wait() and post() operations. Used to control access to a resource. Signaling (1/0) binary semaphore.
- 3. Binary Semaphore**
A special case of a semaphore with a value of 1. Operates as a mutual exclusion between two threads.
- 2. Condition Broadcast**
Wakes up ALL waiting threads for a condition notification.
- 4. Condition Broadcast**
Wakes ALL waiting threads for a condition notification.

BE CAREFUL WITH SEMAPHORES - CAN LEAD TO DEADLOCK!

OPERATING SYSTEMS: ADVANCED SYNCRORIZATION CONDITION VARIABLES & DEADLOCK MITIGATION

DEADLOCK HANDLING



Deadlock: Four necessary conditions: conditions.

1. Deadlock Prevention

Design to avoid one of the four conditions. Example: Resource allocation (e.g., preventing a thread to prevent threads).

2. Deadlock Avoidance

For one's Algorithm

4. Deadlock Avoidance

Use algorithm (e.g., Banker's Algorithm) to grant resources safely.

5. Livelock Detection &

Threads continuously holding resources but making no progress.

1. Deadlock Four necessary conditions: conditions.

Break one Condition. Hold & Wait & Wait. No Preempt circular wait.

Deadlock Avoidance to dynamically to terminate/prevent the set.

Allow deadlock to occur, then detect & detect it and attempt to fix it.

Avoid continuously changing state. Stateful deadlock.

5. Livelock (Starvation)

CVS MUST BE USED WITH A LOCK!